

# Siyuan Gong

📍 Champaign, IL    ✉ siyuan3@illinois.edu    ☎ +1 312-888-5556    🌐 <https://gooosy.github.io/>    🐙 Gooosy

## Education

### University of Illinois at Urbana Champaign

Sept 2023 – Present

*Bachelor of Science in Computer Engineering*

- GPA: 4.0/4.0

### Zhejiang University

Sept 2023 – Present

*Bachelor of Engineering in Electrical and Computer Engineering*

- GPA: 3.90/4.0, Rank: 3/70

## Selective Coursework

- CS 225: Data Structures (A+)
- ECE 391: Computer Systems Engineering (A+)
- ECE 220: Computer Systems & Programming (A+)
- ECE 408: Applied Parallel Programming (A+)

## Honors and Awards

### National Scholarship

*Ministry of Education of People's Republic of China, 2024, 2025*

### Dean's List

*University of Illinois at Urbana Champaign, 2024, 2025*

### Meritorious Winner in Mathematical Contest in Modeling

*Consortium for Mathematics and its Applications, 2024*

### First Prize Scholarship for Academic Excellence

*Zhejiang University-University of Illinois Urbana-Champaign Institute, 2024*

### Gold Medal in iGEM

*International Genetically Engineered Machine (iGEM) Foundation, 2024*

## Research

### Graph-Based In-Context Learning Enhancement

Hangzhou, China

*Advisors: Yifei Sun, Yang Yang, Zhejiang University*

Jun 2024 – Feb 2025

- Conducted literature review, contributed to ideation, and helped refine the model design.
- Designed and implemented experiments: leveraged graph structures and PageRank-based example selection with multi-level encoding (node, path, subgraph), achieving a 2.74% improvement over SOTA baselines.
- Drafted and revised the paper, with emphasis on the experimental methodology and results.

📄 [Paper](#) [↗](#)

## Competition

### International Genetically Engineered Machine Competition

Hangzhou, China

*Team: iZJU-China*

Apr 2024 – Oct 2024

- Constructed a CNN to detect binary dot matrices in yeast images captured under microscope, facilitating the batch processing of biological images.
- Developed a differential equation model to describe the relationship between yeast genes, enabling prediction of potential product expression levels to assist biological experiments.
- Utilized MATLAB to visualize the results for differential equations and create plots of experimental outcomes.
- Wrote Python scripts to automatically handle batch processing tasks.

🌐 [Project Website](#) [↗](#)    🏆 [Gold Medal Certificate](#) [↗](#)

## Mathematical Contest in Modeling

Teammates: Wenjuan Lin, Xinyi Zheng

Hangzhou, China

Feb 2024 – Feb 2024

- Developed the Sea Lamprey Population Model using Fisher's Principle, the Leslie Matrix, and the Lotka-Volterra Model to simulate the dynamics of sea lamprey populations and their impact on ecosystems.
- Utilized MATLAB to stimulate difference equations and visualize the results of the mathematical model through various plots.
- Wrote the paper presenting the final results, addressing how sex ratio variation influences both sea lamprey populations and the surrounding ecosystem.

 [Project Report](#)  [Meritorious Winner Certificate](#) 

## Work

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### Undergraduate Teaching Assistant for ECE 120 Introduction to Computing

Course Instructors: Lin Qiu, Zuozhu Liu, and Ujjal Bhowmik

Haining, China

Sep 2024 – Jan 2025

 [Course Wiki](#) 

### Undergraduate Teaching Assistant for ECE 408 Applied Parallel Programming

Course Instructors: Volodymyr Kindratenko

Illinois, United States

Jan 2026 – Present

## Course Project

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### RISC-V Operating System Kernel

Aug 2025 – Dec 2025

- Implemented a RISC-V operating system kernel on QEMU with interrupt/exception handling, threads, and device drivers in C and assembly.
- Developed virtual memory, system calls, ELF loader, and a simple file system supporting user-mode processes.

### GPT-2 CUDA Kernel Optimizations

Aug 2025 – Dec 2025

- Optimized CUDA kernels for GPT-2 inference on NVIDIA A40 using Nsight Compute and Nsight Systems profiling.
- Applied warp-level reduction, shared-memory tiling, and split-K parallelism, achieving 20× speedup over baseline.

 [Project Report](#) 

### Collaborative Expense Management System

Aug 2025 – Dec 2025

- Built an expense management platform with MySQL on GCP, supporting CRUD, keyword search, multi-currency conversion, and group budgeting.
- Implemented database triggers to maintain group balance consistency and stored procedures with SERIALizable transactions for atomic debt settlement.

 [Project Repo](#) 